HAND & UPPER LIMB · NEET-SS / FET / INI-SS

Congenital hand

1. Embryology

- **Upper limb bud appears at about day 26-28 of gestation**; **limb development is complete by about week 8** (most congenital differences arise between **weeks 4 and 8**).
- **Signalling centres (three axes):**

Axis	Signalling centre	Molecule	Defects
Proximal-distal	**Apical ectodermal ridge (AER)**	**Fibroblast growth factors (FGF-4, FGF-8)**	**Transverse deficiencies (amputations)**, **symbrachydactyly**
Anterior-posterior (radial-ulnar)	**Zone of polarising activity (ZPA)** (posterior mesoderm)	**Sonic hedgehog (SHH)**	**Mirror hand (ulnar dimelia: SHH duplication)**, **polydactyly**, **radial/ulnar deficiencies**
Dorsal-ventral	**Dorsal ectoderm** (**Wnt-7a**) / ventral ectoderm (**engrailed-1 (EN-1)** suppresses Wnt-7a)	**Wnt-7a**, **LMX1B** (dorsal mesoderm)	**Nail-patella syndrome (LMX1B: absent/hypoplastic nails, absent patella)**, **dorsal dimelia (palmar nails)**

- **Digit separation: apoptosis** of interdigital tissue (**bone morphogenetic proteins (BMPs)**) at about **weeks 6-8**: failure = **syndactyly**.
- **HOX genes** (HOXA13, HOXD13: **synpolydactyly**).

2. Classification

- **Swanson (IFSSH, 1976) classification (historical, 7 groups):**
  - **Failure of formation** (transverse: congenital amputations; longitudinal: radial/ulnar/central deficiencies),
  - **Failure of differentiation (separation)** (syndactyly, camptodactyly, clinodactyly, arthrogryposis, synostosis),
  - **Duplication** (polydactyly, mirror hand),
  - **Overgrowth** (macrodactyly),
  - **Undergrowth** (thumb hypoplasia, brachydactyly),
  - **Constriction band syndrome**,
  - **Generalised skeletal abnormalities**.
- **OMT classification (Oberg-Manske-Tonkin; adopted by IFSSH 2014)**: **Malformations** (abnormal axis formation/differentiation: **proximal-distal, radial-ulnar, dorsal-ventral, unspecified axis**), **Deformations** (constriction ring), **Dysplasias** (hypertrophy: macrodactyly; tumorous conditions), **Syndromes**.

3. Syndactyly

- **Commonest congenital hand anomaly (about 1 in 2,000-2,500 live births)**; **boys 2:1**; **bilateral about 50%**; **familial (autosomal dominant with variable penetrance) in about 10-40%**.
- **Commonest site: third web space (middle-ring fingers) (about 50%)**, then 4th, 2nd, 1st (rare).
- **Classification:**
  - **Simple** (skin/soft tissue only) vs **complex** (**bony fusion: side-to-side, usually distal phalanges**),
  - **Complete** (extends to the fingertip) vs **incomplete**,
  - **Complicated** (**abnormal bones: e.g. Apert syndrome, synpolydactyly**).
- **Associated syndromes:**

  - **Apert syndrome (acrocephalosyndactyly; FGFR2 mutation)**: **craniosynostosis, mid-face hypoplasia, complex "mitten hand" syndactyly (rosebud hand)**, **symphalangism**, **radial clinodactyly of the thumb (delta phalanx)**,
  - **Poland syndrome**: **unilateral absence of the sternocostal head of pectoralis major**, **ipsilateral symbrachydactyly (short webbed fingers: hypoplastic middle phalanges)**, **breast/nipple hypoplasia**,
  - **Constriction band syndrome (acrosyndactyly: fenestrated distal fusion with proximal sinus tracts)**,
  - **Carpenter syndrome**, **Pfeiffer syndrome**.
- **Timing of release:**
  - **Border digits (thumb-index, ring-little) first and earlier (about 6-12 months)**: **unequal lengths cause progressive tethering/angular deformity of the longer digit**,
  - **Central digits (middle-ring): about 12-18 months** (before school age),
  - **Release only one side of a digit at a time** (**never both sides of a finger simultaneously: risk of vascular compromise**),
  - Complex syndactyly: earlier staged release.
- **Technique: dorsal rectangular or pentagonal flap to reconstruct the web commissure** (**the commissure slopes from dorsal-proximal to palmar-distal at about 45 degrees**), **interdigitating zig-zag flaps**, **full-thickness skin grafts (groin/antecubital)** for skin deficit (**syndactylised digits share skin: almost always need grafts**); **Buck-Gramcko pulp plasty** for fingertip separation (nail folds).
- **Complications: web creep (commonest: distal migration of the web: especially with split-thickness grafts or early surgery)**, **flexion contractures (scar)**, **graft loss**, **nail deformity**, **vascular compromise**, **angular deformity**.

4. Polydactyly

- **Postaxial (ulnar) polydactyly: commonest overall** (**about 10x more common in individuals of African descent**; **autosomal dominant**); **Temtamy and McKusick: type A (well-formed digit articulating with the fifth/sixth metacarpal)**, **type B (rudimentary, pedunculated skin tag ("nubbin"))**. **In Caucasians, postaxial polydactyly suggests a syndrome** (e.g. **Ellis-van Creveld (chondroectodermal dysplasia)**, **Bardet-Biedl**, **trisomy 13**): **genetic evaluation**. **Type B: suture ligation in the nursery (risk of a residual bump/neuroma) or surgical excision**; **type A: surgical excision with reconstruction of the collateral ligament/hypothenar muscle (ADM) insertion** at about **9-12 months**.
- **Preaxial (radial) polydactyly = thumb duplication:**
  - **Commonest in Asians and Caucasians among radial anomalies**; **usually sporadic, unilateral**.
  - **Wassel classification (level of duplication; odd numbers = bifid (partial), even numbers = complete duplication):**
  - **I** bifid distal phalanx,
  - **II** duplicated distal phalanx,
  - **III** bifid proximal phalanx,
  - **IV duplicated proximal phalanx (commonest: about 45%)**,
  - **V** bifid metacarpal,
  - **VI** duplicated metacarpal,
  - **VII triphalangeal thumb component** (associated with **Holt-Oram syndrome, Fanconi anaemia (VII + radial deficiency)**, autosomal dominant triphalangeal thumb).
  - **Wassel-Flatt** (modified) and **Rotterdam classification** (more inclusive).
  - **Treatment (about 12 months): excise the hypoplastic (usually radial) thumb**, **preserve and reconstruct the radial collateral**

ligament (**reattach to the retained thumb with a periosteal sleeve**), **reattach the thenar muscles (APB) to the retained thumb**, **correct the axis (osteotomy)** and **centralise the tendons (FPL/EPL eccentric insertions)**; **Bilhaut-Cloquet procedure** (**central wedge resection and combination of both halves**: equal-sized hypoplastic thumbs: Wassel I/II; risk of **nail deformity and physeal mismatch/stiffness**).

- **Complications: Z-deformity (zigzag: angular deformity at the IPJ/MCPJ)** (commonest late problem: from unbalanced tendons/collateral ligaments), **instability**, **stiffness**, residual nail deformity.

- **Central polydactyly** (index, middle, ring duplications): often with **syndactyly (synpolydactyly: HOXD13)**; complex reconstruction.
- **Mirror hand (ulnar dimelia)**: **duplication of ulnar side (two ulnae, no radius, multiple fingers (7-8), no thumb)**: **ZPA duplication (SHH)**; **pollicisation of one digit + excision**.

5. Radial longitudinal deficiency (radial club hand)

- **Spectrum** of hypoplasia of the radius and pre-axial (thumb) structures: **about 1 in 30,000-100,000**; **bilateral about 50%**; **thumb always involved** (hypoplasia/absence).
- **Bayne and Klug classification (radius):**
  - **Type 0: normal radius length** (carpal deficiency/thumb hypoplasia only) (added later),
  - **Type I: short distal radius** (delayed distal radial physis),
  - **Type II: hypoplastic radius** (miniature, both physes),
  - **Type III: partial absence** (usually distal: commonest),
  - **Type IV: complete absence of the radius (commonest overall: about 50%)**.
  - (**James/Manske modified: type N** (thumb only), **type 0** (carpal anomalies with normal radius length)).

- **Associated syndromes (screen all patients):**
  - **VACTERL** (Vertebral, Anal atresia, Cardiac, TracheoEsophageal fistula, Renal, Limb anomalies),
  - **Holt-Oram syndrome** (**cardiac septal defects: ASD/VSD; TBX5, autosomal dominant**),
  - **TAR syndrome** (**thrombocytopenia-absent radius**: **thumbs PRESENT** (unique); platelets improve after the first year; autosomal recessive (RBM8A)),
  - **Fanconi anaemia** (**aplastic anaemia (develops about age 6-7: chromosomal breakage test: DEB/mitomycin C)**; needs **bone marrow transplant**; screening is essential),
  - **Trisomy 13/18**.
  - **Investigations: CBC/platelets, echocardiogram, renal ultrasound, spine radiographs, chromosomal breakage test (Fanconi)**.
- **Clinical: radially deviated, short forearm**, **ulna bowed (towards the radius)**, **thumb hypoplasia/absence**, **stiff elbow (elbow extension contracture: contraindicates centralisation, as the radially deviated wrist allows the hand to reach the mouth)**; **radial nerve absence distal to the elbow (superficial radial nerve replaced by the median nerve)**; **radial artery absent**.
- **Treatment:**
  - **Early: stretching and splinting** (from birth).
  - **Types I-II (mild)**: stretching/splinting; **radial lengthening**.
  - **Types III-IV**: **centralisation (placing the carpus on the distal ulna)** or **radialisation (Buck-Gramcko: carpus over-corrected onto the ulna with transfer of radial wrist extensors to the ulnar side)** at **about 6-12 months** (after soft-tissue distraction with an external fixator), followed by **pollicisation (of the index finger) for an absent/non-reconstructable thumb**; **ulnar osteotomy** for bowing.

- **Contraindications to centralisation: stiff elbow (extension contracture: patient needs radial deviation to reach the mouth)**, **adults who have adapted well**, **severe associated medical conditions (short life expectancy)**.
- **Recurrence of radial deviation (commonest complication)**, **ulnar physeal damage (growth arrest)**, stiffness.

6. Thumb hypoplasia

- **Blauth classification (modified by Manske):**
  - **Type I: minor generalised hypoplasia** (all elements present, small): no treatment,
  - **Type II: thenar muscle hypoplasia, first web space narrowing, ulnar collateral ligament (UCL) instability of the MCPJ**: **opponensplasty (Huber: ADM transfer; or FDS ring)**, **first web deepening (4-flap Z-plasty)**, **UCL reconstruction**,
  - **Type III: extrinsic tendon abnormalities (pollex abductus: FPL connection to EPL)**, **hypoplastic metacarpal**: **IIIA: stable CMCJ (basal joint present)**: **reconstruct (like type II + extrinsic tendon correction)**; **IIIB: unstable CMCJ (absent base of metacarpal)**: **pollicisation** (Manske: **critical distinction: presence of a stable CMCJ**),
  - **Type IV: pouce flottant ("floating thumb"): rudimentary thumb attached by a skin bridge**: **pollicisation**,
  - **Type V: absent thumb**: **pollicisation**.
- **Pollicisation (Buck-Gramcko technique)**: **index finger transferred to the thumb position**: **rotated about 120-160 degrees (pronation)**, **abducted about 40 degrees**, **shortened** (**index MCPJ becomes the new CMCJ; metacarpal head becomes the trapezium; PIPJ becomes the MCPJ**); **first dorsal interosseous becomes APB/abductor**, **first palmar interosseous becomes adductor**, **EIP becomes EPL**, **EDC (index) becomes APL**; **at about 12-18 months**.

7. Ulnar longitudinal deficiency

- **Less common than radial (about 1:4-10 relative frequency)**; **not associated with systemic syndromes** (unlike radial deficiency) but **associated with other musculoskeletal anomalies (PFFD, fibular hemimelia, scoliosis)** and **hand anomalies (ulnar digits absent, syndactyly)**.
- **Bayne classification: I hypoplastic ulna**, **II partial aplasia**, **III total aplasia**, **IV radiohumeral synostosis**.
- **Elbow instability or synostosis** (vs radial deficiency: wrist is relatively stable).
- **Treatment:** focus on **hand function (syndactyly release, thumb reconstruction, first web)**; **excision of the ulnar anlage** (fibrocartilaginous band that tethers the radius: progressive bowing/radial head dislocation), **one-bone forearm** for severe instability.

8. Central deficiency (cleft hand) vs symbrachydactyly

Feature	Cleft hand (typical; central longitudinal deficiency)	Symbrachydactyly (atypical cleft)
Shape	**V-shaped central cleft** (missing middle ray)	**U-shaped**; **short digits with nubbins (rudimentary nails)**
Inheritance	**Autosomal dominant; often bilateral**; **feet involved (split hand-foot malformation: SHFM)**	**Sporadic, unilateral**
Associations	**EEC syndrome (ectrodactyly-ectodermal dysplasia-clefting)**, syndactyly of the first web	**Poland syndrome**

Feature	Cleft hand (typical; central longitudinal deficiency)	Symbrachydactyly (atypical cleft)
Mechanism	**AER (central) failure**	**AER/mesoderm (proximal-distal) failure: transverse spectrum**
Function	**Often excellent ("functional triumph, aesthetic disaster" (Flatt))**	Variable
Treatment	**Cleft closure (Snow-Littler, Ueba, Miura)** + **first web release** (**syndactyly of the first web: release first**); **transposition of the index ray** to the third metacarpal position	**Non-vascularised toe phalanx transfer** (**under 18 months: physis survives**), **free toe transfer**, **distraction lengthening**, **web deepening**

9. Camptodactyly and clinodactyly

Camptodactyly

- **Non-traumatic flexion contracture of the PIPJ** (**little finger commonest**; **bilateral about 2/3**); **infantile (equal sexes)** and **adolescent (girls; progressive during growth spurt)** forms.
- **Cause: abnormal lumbrical insertion (commonest)**, **FDS contracture/abnormal origin**, **skin deficiency**, volar plate contracture, **abnormal insertion of the lumbrical/interosseous into the FDS or capsule**.
- **Radiographs (late): flattening (notching) of the neck of the proximal phalanx, flattened/indented base of the middle phalanx, volar subluxation**.
- **Treatment: splinting and stretching (mainstay; most effective under about 3 years and in flexible deformity)**; **surgery (FDS tenotomy/lengthening, abnormal lumbrical release, volar plate**

release, skin flaps/grafts) for **contractures over 30-60 degrees with functional impairment or progression despite splinting**; **outcomes are unpredictable** (stiffness, loss of flexion).

Clinodactyly

- **Angular deformity in the radioulnar (coronal) plane**, **little finger (radial deviation at the DIPJ) commonest**; **middle phalanx**: **trapezoidal or triangular (delta phalanx: longitudinal epiphyseal bracket (C-shaped physis))**.
- **Associations: Down syndrome (about 35-80%)**, Klinefelter, Turner, **Rubinstein-Taybi (broad thumbs, delta phalanx)**, **Apert**.
- **Treatment: observation (most: cosmetic)**; **surgery if over 30 degrees with functional impairment/overlapping**: **opening-wedge, closing-wedge or reverse-wedge osteotomy**; **physiolysis (Vickers: excision of the longitudinal bracket with fat interposition)** in **young children (under 6 years: allows growth to correct)**.
- **Kirner deformity**: **palmar-radial curvature of the distal phalanx of the little finger** (**girls 8-14 years**): observation/osteotomy.

10. Other conditions

- **Trigger thumb (paediatric): not truly congenital (develops after birth; usually noticed at 1-3 years)**; **fixed flexion of the IPJ** with a **Notta node (nodule in the FPL at the A1 pulley)**; **about 30% resolve spontaneously (more under 1 year)**; **A1 pulley release** if persistent beyond about **3 years** (or earlier if fixed) (**the radial digital nerve is close and superficial: at risk**).
- **Trigger finger (paediatric):** rarer; **anatomical abnormalities (FDS decussation, abnormal slips, nodules)**: **A1 release alone is often insufficient: excise one slip of FDS ± A3 release**.
- **Congenital clasped thumb: absent/hypoplastic EPB/EPL** (thumb flexed across the palm); **splinting (3-6 months)**; **tendon transfer**

(EIP to EPL) and **web reconstruction** if it fails; associated with **arthrogryposis** (**complex clasped thumb**).

- **Constriction ring (amniotic band) syndrome: circumferential constriction rings**, **acrosyndactyly (fenestrated distal syndactyly)**, **amputations**, **lymphoedema distal to the ring**; **not genetic (sporadic)**; **urgent release for vascular compromise (neonatal emergency)**; **staged Z-plasty/W-plasty release (half the circumference at a time traditionally; single-stage now common)**.
- **Macrodactyly: true overgrowth of all tissues of a digit** (**PIK3CA-related overgrowth spectrum (PROS)**; **lipofibromatous hamartoma of the median nerve** ("nerve territory-oriented macrodactyly": index/middle)); **static vs progressive**; **epiphysiodesis, debulking, ray amputation**; differentials: **Proteus syndrome**, **neurofibromatosis**, **Klippel-Trenaunay**, **Ollier/Maffucci**.
- **Arthrogryposis multiplex congenita (amyoplasia commonest)**: **internally rotated shoulders, extended elbows, flexed and ulnarly deviated wrists, thumb-in-palm**, **symmetrical limb involvement, absent skin creases**, **normal intelligence and sensation**; **early stretching/serial casting**; **elbow flexion (passive flexion first: posterior capsulotomy + triceps lengthening; then tendon transfers (triceps to biceps, pectoralis, latissimus))**; **wrist: dorsal closing-wedge carpal osteotomy**.
- **Congenital radioulnar synostosis: proximal (commonest)**, **fixed pronation**, **often bilateral**; **AD inheritance in some; associated with sex chromosome anomalies (XXY)**; **no rotation (no supination)**; **observation if pronation under about 60 degrees** (**compensated by shoulder/wrist**); **derotational osteotomy through the synostosis to about 10-20 degrees pronation (dominant) and neutral/slight supination (non-dominant)** if severe (over about 60 degrees); **compartment syndrome and**

vascular compromise risk with large rotations; **resection of the synostosis fails (re-forms)**.

- **Madelung deformity: volar-ulnar tethering of the distal radial physis (Vickers ligament: abnormal radiolunate ligament)**: **volar and ulnar tilt of the distal radius, dorsally prominent ulna**, **adolescent girls, bilateral**; **Léri-Weill dyschondrosteosis (SHOX gene)**; **Vickers ligament release with physiolysis (younger)**, **dome osteotomy of the radius ± ulnar shortening** (older).
- **Congenital radial head dislocation: usually posterior**, bilateral, **convex radial head, hypoplastic capitellum** (vs traumatic: concave head, normal capitellum); **associated with syndromes (nail-patella, Ehlers-Danlos)**; **observe; excise the radial head at maturity** if painful.

> Exam pearl: **AER (FGF) = proximal-distal (transverse deficiency); ZPA (SHH) = radial-ulnar (polydactyly, mirror hand); Wnt-7a/LMX1B = dorsal-ventral (nail-patella). OMT classification replaced Swanson.**

> Exam pearl: **Syndactyly: commonest; 3rd web; border digits first (6-12 months); never both sides of one digit; dorsal flap commissure + FTSG; web creep. Apert (FGFR2) = complex mitten hand; Poland = pectoralis head + symbrachydactyly.**

> Exam pearl: **Thumb duplication: Wassel IV commonest; excise radial thumb, keep RCL and thenar insertion; Bilhaut-Cloquet for equal halves; Z-deformity late. Postaxial polydactyly in Caucasians = syndrome (Ellis-van Creveld).**

Exam pearl: **Radial deficiency: screen for VACTERL, Holt-Oram (heart), TAR (thumbs present), Fanconi (chromosomal breakage). Bayne IV commonest. Centralisation/radialisation contraindicated with a stiff elbow. Blauth IIIA reconstruct, IIIB/IV/V pollicise (stable CMCJ is the key). Pollicisation: index rotated about 120-160 degrees; MCPJ becomes CMCJ.**

Exam pearl: **Cleft hand (V, bilateral, feet, AD) vs symbrachydactyly (U, unilateral, nubbins, Poland). Camptodactyly: PIPJ little finger (lumbrical), splint. Clinodactyly: delta phalanx, physiolysis under 6. Paediatric trigger thumb: Notta node, release after 3 years (radial digital nerve at risk). Synostosis: osteotomy, not resection. Madelung: Vickers ligament, SHOX.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.